THE PH COVERINGS FORM A TOPOLOGY: IDENTITY, COMPOSITION, BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Base change of locally finite type morphisms. Morphisms of Spaces, lemma base-change-finite-type (cross-chapter, named). X' x_X X_i -> X' stays locally of finite type.
- Scheme-level standard ph refinement. Topologies, lemma refine-by-standard-ph (cross-chapter, named). The doubled family in (2) is refined by a standard ph covering.

Source reference. Original reference 03XH: focused-seeds/geometry/S9-topo-ph/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0DBF: focused-seeds/geometry/S9-topo-ph/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S.*

(1) *If $X' \to X$ is an isomorphism then $\{X' \to X\}$ is a ph covering of X.*
(2) *If $\{X_i \to X\}_{i \in I}$ is a ph covering and for each i we have a ph covering $\{X_{ij} \to X_i\}_{j \in J_i}$, then $\{X_{ij} \to X\}_{i \in I, j \in J_i}$ is a ph covering.*
(3) *If $\{X_i \to X\}_{i \in I}$ is a ph covering and $X' \to X$ is a morphism of algebraic spaces then $\{X' \times_X X_i \to X'\}_{i \in I}$ is a ph covering.*

Proof. Part (1) is clear. Consider $g : X' \to X$ and $\{X_i \to X\}_{i \in I}$ a ph covering as in (3). By Morphisms of Spaces, Lemma 03XH the morphisms $X' \times_X X_i \to X'$ are locally of finite type. If $h' : Z \to X'$ is a morphism from an affine scheme towards X', then set $h = g \circ h' : Z \to X$. The assumption on $\{X_i \to X\}_{i \in I}$ means there exists a standard ph covering $\{Z_j \to Z\}_{j=1,\ldots,n}$ and morphisms $Z_j \to X_{i(j)}$ covering h for certain $i(j) \in I$. By the universal property of the fibre product we obtain morphisms $Z_j \to X' \times_X X_{i(j)}$ over h' also. Hence $\{X' \times_X X_i \to X'\}_{i \in I}$ is a ph covering. This proves (3).

Let $\{X_i \to X\}_{i \in I}$ and $\{X_{ij} \to X_i\}_{j \in J_i}$ be as in (2). Let $h : Z \to X$ be a morphism from an affine scheme towards X. By assumption there exists a standard

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ph covering $\{Z_j \to Z\}_{j=1,\ldots,n}$ and morphisms $h_j : Z_j \to X_{i(j)}$ covering h for some indices $i(j) \in I$. By assumption there exist standard ph coverings $\{Z_{j,l} \to Z_j\}_{l=1,\ldots,n(j)}$ and morphisms $Z_{j,l} \to X_{i(j)j(l)}$ covering h_j for some indices $j(l) \in J_{i(j)}$. By Topologies, Lemma 0DBF the family $\{Z_{j,l} \to Z\}$ can be refined by a standard ph covering. Hence we conclude that $\{X_{ij} \to X\}_{i\in I, j\in J_i}$ is a ph covering. $\square$